FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

FULL TEST – V

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 186**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 05, 19 – 23, 37 - 41)*** *contains 15 multiple choice questions which have* ***one correct*** *answer. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (06 – 13, 24 – 31, 42 - 49) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

2. ***Section-C (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

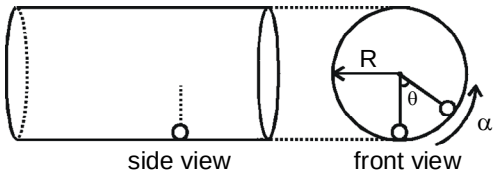
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A small sphere of mass m and radius r is placed inside a horizontal hollow cylinder of radius R. Now cylinder rotating with some angular acceleration which slowly increases from zero to certain value α.



(A) Maximum angle θ formed by line joining centre will be $\theta = \sin^{-1}\left(\frac{2\alpha R}{5g}\right)$

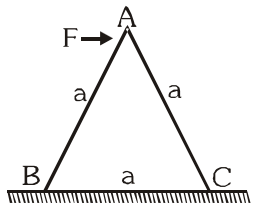
(B) Angular velocity of sphere will be $\omega' = \frac{\omega r}{R}$ where ω is angular velocity of cylinder

(C) Sphere will keep rotating at lowest position

(D) K.E. of sphere becomes constant when it reaches maximum height.

2. You are holding a bottle of bubbled water inside a car moving forward. When the driver applies the brakes:
(A) Bubbles in the middle of the liquid will start to move forward with respect to the bottle.
(B) Bubbles will start to move backward with respect to the bottle.
(C) Bubbles will stay at the same horizontal location in the water
(D) Depending on the speed of the car, bubbles might move forward or backward.

3. A horizontal force F is applied at the top of an equilateral triangular block having mass m. The minimum coefficient of friction required to topple the block before translation will be

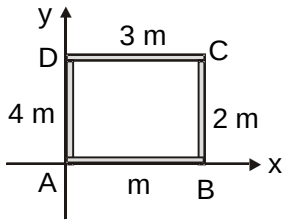


(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{3}}$ (D) None of these

Space for Rough work

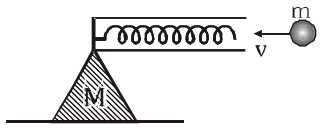
4. Four rods of equal lengths AB, BC, CD and DA have mass m, 2m, 3m and 4m respectively are place in x-y plane (see figure) given AB=BC=CD=DA=4 meters. The centre of mass of the system will satisfy (x and y are in meters)



(A) $x > 2,\ y > 2$ (B) $x > 2,\ y < 2$

(C) $x < 2,\ y < 2$ (D) $x < 2,\ y > 2$

5. A ball of mass m = 6 kg is shot with speed v = 20 m/s into a barrel of spring gun of mass M = 24 kg initially at rest. All surfaces are smooth & spring is massless. Speed of gun after ball stops relative to gun is



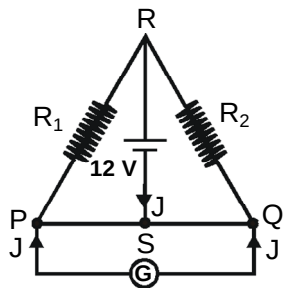
(A) 2 m/s (B) 4 m/s

(C) 6 m/s (D) 8 m/s

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

6. In the circuit shown $R_1 = R_2 = 10\,\Omega$ and resistance per unit length of wire PQ = $1\,\Omega$/cm and length PQ = 10 cm. If R_2 is made $20\,\Omega$ then to get zero deflection in galvanometer. S is midpoint of wire PQ



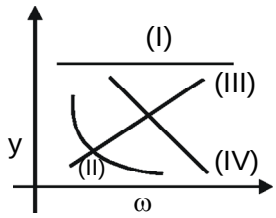
(A) The jockey at P can be moved towards right 2 cm

(B) The jockey at Q can be moved towards left 2cm

(C) The jockey at S can be moved towards left a distance 5/3 m

(D) The jockey at all position fixed and R_1 should be made $20\,\Omega$

7. In a series LCR circuit, different physical quantities vary with frequency ω. Which of the following curves represent correct frequency variation of the corresponding quantity?



(A) Curve I for R

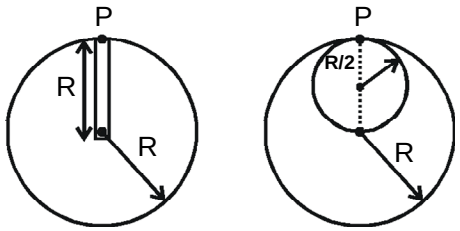
(B) Curve II for current I

(C) Curve III for X_L

(D) Curve IV for X_C

Space for Rough work

8. Consider two solid spherical asteroid of uniform density of mass M and radius R. In one asteroid a tunnel of very small size of depth R is bored to the centre and in other asteroid a spherical cavity of radius R/2 is made as shown in the figure. Now, identical particles of mass m dropped into the cavities of both asteroids from the top most point P. If force experienced by particle is F_I and F_{II}



Asteroid - I Asteroid - II

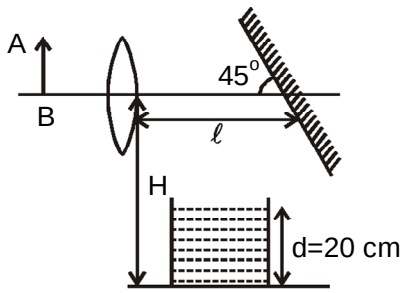
(A) The ratio of F_1 / F_{II} is equal to 2x/R
(B) The ratio of F_I / F_{II} is equal to x/R
(C) The ratio of T_I / T_{II} is equal to $\pi/4$
(D) The ratio of T_I / T_{II} is equal to $\pi/2$

9. Electric field in a region is given as $\vec{E} = (10 - 5x)\hat{i}$. A charge particle of mass 5kg and charge $Q(= 1C)$ is situated at origin and free to move in given electric field. Then choose the correct options (Neglect gravity):-
(A) Motion of charge particle is Oscillatory
(B) Maximum displacement of charge particle from origin in 4 SI units
(C) Maximum velocity gain by charge particle is 2 SI units
(D) The position of charge particle, when velocity gained by particle is maximum, is 2 SI units

10. A proton and an electron are moving with the same de-Broglie wavelength (consider the non-relativistic case). Then
(A) In a magnetic field both the particles describe circles of equal radius
(B) Both the particles have the same momentum
(C) The speed of the proton and the electron are in the ratio m_e / m_p, where m_e is the electron mass and m_p, the proton mas
(D) The product of mas and kinetic energy is the same for both particles

Space for Rough work

11. A linear object AB is at a distance of 36 cm from an equal convex lens of focal length 30 cm. In front of lens there is a plane mirror which is inclined at an angle 45° with the principal axis of the lens at a distance $l = 1$ from the mirror as shown in figure. A container with water layer d = 20 cm is placed as shown in the figure. Then choose the **CORRECT** statement(s). Take the refractive index of water is $\frac{4}{3}$



(A) After reflection from the mirror the image of AB will be parallel to principal axis of the lens.
(B) After reflection from the mirror the image of AB will be perpendicular to principal axis of the lens.
(C) The value of H for which the sharp image of AB can be obtained at the bottom of the container is 80 cm.
(D) The value of H for which the sharp image of AB can be obtained at the bottom of the container is 85 cm.

12. Two identical objects A and B $(\text{emissivity } e_A \text{ and } e_B; e_A \neq e_B)$ are placed in an enclosure. Temperature of body A, B and enclosure are same and constant : -
(A) Heat emitted by body A is not equal to heat emitted by body B
(B) Heat absorbed by body A is not equal to heat absorbed by body B
(C) If $e_A > e_B$, body A absorbs more heat than body B
(D) If $e_A < e_B$, body A emits less heat than body B

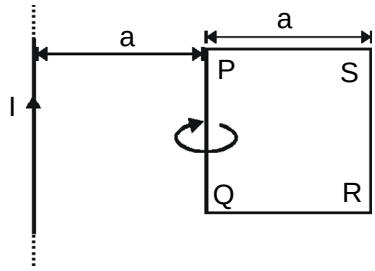
13. ${}_{92}U^{235}$ is 'α' active. Then in a large quantity of the element:-
(A) The probability of a nucleus disintegrating during one second is lower in the first half life and greater in the fifth life
(B) The probability of a nuelcues disintegrating during one second remains constant for all time.
(C) Quite an appreciable quantity of U^{235} will remain even the average life
(D) The energy of the emitted 'α' particle is less than the disintegration energy of the U^{235} nucleus.

Space for Rough work

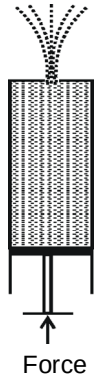
SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

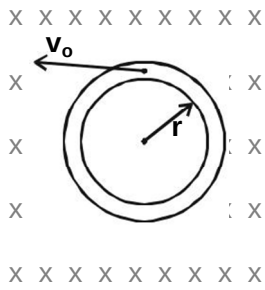
14. A square frame of resistance $\ell n(2)\Omega$ and side a = 20 cm and a long straight wire carrying a current I = 10 amp are located in the same plane. The frame is rotated through an angle of 120° about the side PQ. The amount of charge flown through the loop during this time is $q\times10^{-7}$ coulomb, find the value q.



15. A syringe is filled with water up to volume 20 cm^3. The area of cross – section of the cylinder is 5 cm^2. The syringe is held vertically and its 90 gm piston is pushed upward by external agent with constant speed. A water beam coming out of the small nozzle (hole area 1 mm^2). Neglecting friction and viscous nature of water, find the work done by the agent [in 10^{-2}] in fully emptying the syringe. (take g = 10 m/s^2)

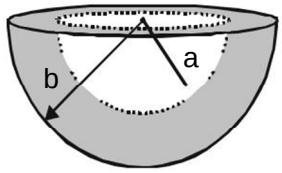


16. A horizontal tube of small thickness having inner radius r = R/2 is placed in gravity free space. Magnet field of strength B is present perpendicular to plane of circular tube. A charged particle (q. m) inside, the tube is given tangential velocity $v_o = \frac{qBR}{m}$. The work done by friction long after motion is $-\frac{Kq^2B^2R^2}{8m}$. Find the value of K. [inner surface of tube is rough]



Space for Rough work

17. A uniformly charged hemisphere of radius b and charge density ρ has a hemispherical cavity of radius $a\left(a=\frac{b}{2}\right)$ cut from its centre. If the potential at the centre of the cavity is $\frac{n\rho b^2}{16\in_0}$ then n = ?



18. Two sound waves of frequencies 100 Hz and 102 Hz and having same amplitude 'A' are interfering. At a stationary detector, which can detect resultant amplitude grater than or equal to A. So, in a given time interval of 12 seconds, find the total duration in which detector is active.

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. In an aqueous solution of volume 500 ml, when the reaction given below $2Ag^{+} + Cu \rightleftharpoons Cu^{2+} + 2Ag$ reached at equilibrium, the $\left[Cu^{2+}\right]$ was x M. When 500 ml of water is further added, at the equilibrium, the $\left[Cu^{2+}\right]$ will be

(A) x (B) 2x

(C) less than $\frac{x}{2}$ (D) $\frac{3x}{4}$

20. When Sulphur is boiled with Na_2SO_3, a compound (A) is produced. With excess of $AgNO_3$ solution, compound (A) gives a compound (B), which is dissolved in water and produces a black coloured compound (C) containing sulphur. 'C' is

(A) $Ag_2S_2O_3$ (B) Ag_2SO_4

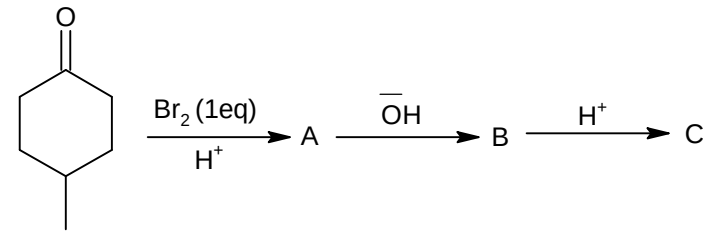
(C) $\left[Ag_2\left(S_2O_3\right)_4\right]^{6-}$ (D) Ag_2S

21. The rate of disintegration of a certain radioactive sample at any instant is 8100 dpm. Fifty minutes later, the rate becomes 2700 dpm. The half life of the radioactive sample will be (Take log2 = 0.3, log3 = 0.5, log5 = 0.7)

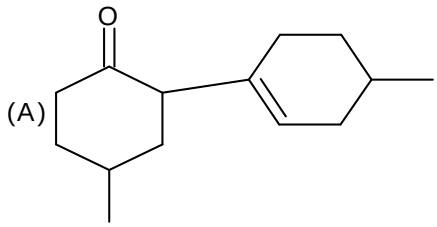
(A) 30 minutes (B) 61 minutes

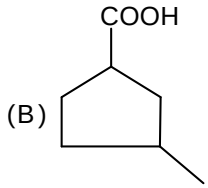
(C) 47.8 minutes (D) 33.3 minutes

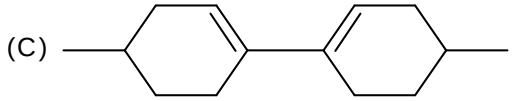
Space for Rough work

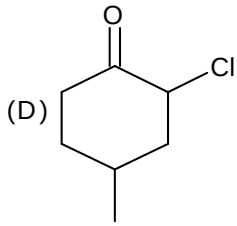
22. 

C is









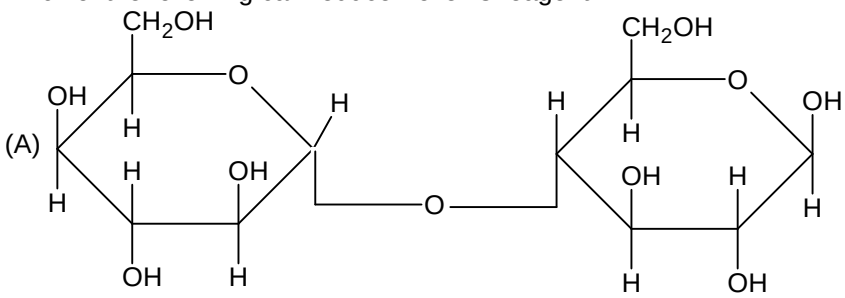
23. The IUPAC name of C_2H_5NHCHO is
(A) N – aldoethanamine
(B) N - formylethanamine
(C) N - aldoaminoethane
(D) N – ethylmethanamide

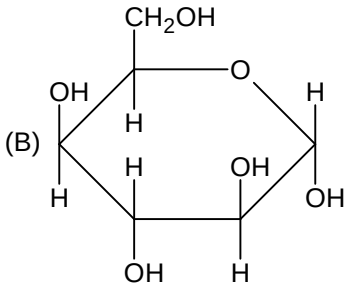
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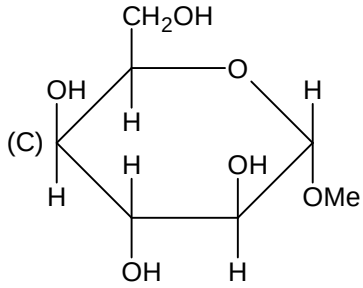
(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

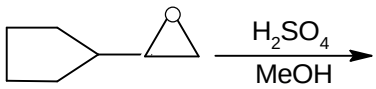
24. Which of the following can reduce Tollen's reagent?

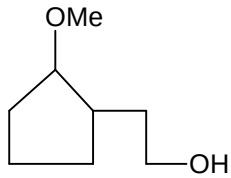
(A)


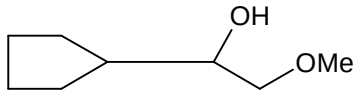
(B)


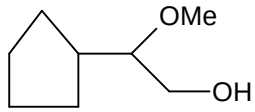
(C)


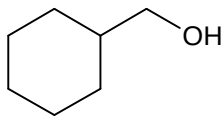
(D) fructose

25.


(A)


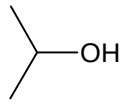
(B)


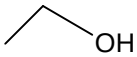
(C)


(D)


Space for Rough work

26. Which of the following will produce red color when treated with $(NH_4)_2\left[Ce(NO_3)_6\right]$?

(A) CH_3OH (B) 

(C)  (D) PhOH

27. Which of the following statement is/are correct?

(A) On adding excess of Ag NO_3 to 1 mole of each $PtCl_4$. 4 NH_3 and $\left[Ni(en)_3\right]Cl_2$, same weight of ppt (AgCl) is produced.

(B) Ma_3b_3 complexes do not show optical activity

(C) Oxidation number of central metal in Zeise's salt is + 2.

(D) $K_4\left[Fe(CN)_6\right]$ & $K_3\left[Fe(CN)_6\right]$ both have sp^3d^2 hybridisation.

28. Which of the following trend is/are correct?

(A) $ClO_2 > SO_2 > CO_2 > P_2O_5$ (acidic nature) (B) Na > Si > Al > Mg (second I. P.)

(C) S > Se > Te > O (electron affinity) (D) $H_3PO_2 > H_3PO_3 > H_3PO_4$ (Ka_1)

29. Which of the following will dissolve in liq. HF without any chemical reaction?

(A) XeF_2 (B) XeF_4

(C) XeF_6 (D) NaH

30.

(1)  $\xrightarrow[(d)]{n\text{ - BuLi}}$ A + B (B does not contain π bond)

(2) $A \xrightarrow{Cl - CH_2O\ Me} C$

Which is/are true for above reactions?

(A) B can be prepared by corey house synthesis

(B) C after hydrogenation on treatment with 1 eq cold conc HI produces an alkyl halide with highest density

(C) C does not decolorizes Br_2 – water

(D) A is organometallic compound

Space for Rough work

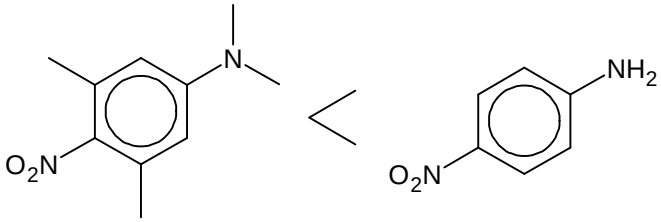
31. Which of the following is/are wrong statement(s)?
(A) The second I.P. is always less than the 1st I.P. of an atom
(B) If the addition of an electron to an ion leads to complete octet, then the process is always exothermic
(C) Chlorine has highest electronegativity
(D) 1st I.P. of N is higher than 1st I.P. of F, because N has half filled subshell

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. An Ester produces only one alcohol on treatment with CH_3MgBr followed by hydrolysis and this ester has minimum molecular weight, find the number of carbon atoms present in ester which satisfies above conditions.

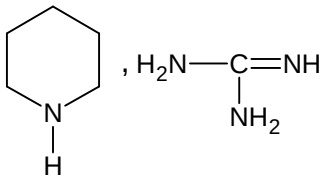
33. How many of the following is/are correct order of dipole moment?
(A) $CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$
(B) $H - F > H - Cl > H - Br > H - I$
(C) $CH_3 - Cl > CH_2Cl_2 > CHCl_3 > CCl_4$
(D) $CH_2 = CH - Cl < CH_3 - CH_2 - Cl$
(E)



(F) $CH_3 - Cl > CH_3 - F > CH_3 - Br$
(G) $H - Cl > H - F > H - Br$

Space for Rough work

34. How many of the following are more basic than aniline?
O – toluidine, O – anisidine, m – anisidine, P – anisidine, P – chloroaniline, m – toluidine,



35. How many of the following carbides on reaction with water will produce a product capable of forming silver mirror, on treatment with Tollen's reagent?
Be_2C, CaC_2, Mg_2C_3, Al_4C_3, CH_4, SiC.

36. The distance between an octahedral and tetrahedral void in FCC lattice would be $\frac{\sqrt{3}a}{b}$. Find the value of b if a is edge length of cube.

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **5 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. The length of focal chord AB of ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ is: $\left(\text{Given } A=\left(\frac{8}{5},\ \frac{3\sqrt{3}}{5}\right)\right)$

(A) $\frac{4}{5}$ (B) $\frac{16}{5}$

(C) $\frac{32}{5}$ (D) $\frac{64}{5}$

38. If $2x-y+1=0$ is a tangent on the parabola which intersect its directrix at (1, 3) and focus is (2, 1), then equation of axis of parabola is

(A) $11x-2y=24$ (B) $11x+2y+24=0$

(C) $2x-11y-24=0$ (D) $2y+11x=24$

39. If vectors $\hat{a}, \hat{b}, \hat{c}$ are in space such that $\hat{a}.\hat{b}=\hat{b}.\hat{c}=\hat{c}.\hat{a}=\frac{1}{2}$, then $\left(\hat{a}\times\hat{b}\right).\left(\hat{a}\times\left(\hat{b}\times\hat{c}\right)\right)$ is equal to

(A) $\frac{1}{\sqrt{2}}$ (B) $-\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2\sqrt{2}}$ (D) $-\frac{1}{2\sqrt{2}}$

40. How many matrices X with entries {0, 1, 2} are there for which sum of diagonal entries of $X.X^T$ is 7?

(A) 560 (B) 562

(C) 504 (D) 540

Space for Rough work

41. If α and β are roots of equation $\frac{3}{4}\sin\left(\frac{\theta}{9}\right) = \sin^3\theta + 3\sin^3\left(\frac{\theta}{3}\right) + 9\sin^3\left(\frac{\theta}{9}\right) + \frac{1}{4\sqrt{2}}$ for $0 < \theta < \frac{\pi}{2}$, then $\tan\alpha + \tan\beta$ is equal to

(A) $2+\sqrt{3}$ (B) $3+\sqrt{3}$

(C) $3-\sqrt{3}$ (D) $2-\sqrt{3}$

(One or More than one correct type)

This section contains **EIGHT** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

42. If volume of a parallelepiped determined by the vectors $\bar{a}$, $\bar{b}$ & $\bar{c}$ is 2, then volume of parallelepiped determined by the vectors

(A) $3(\bar{a}+\bar{b})$, $(\bar{b}+\bar{c})$ and $(\bar{c}+\bar{a})$ is 24 (B) $2(\bar{a}\times\bar{b})$, $3(\bar{b}\times\bar{c})$ and $(\bar{c}\times\bar{a})$ is 24

(C) $(\bar{a}-2\bar{b})$, $(\bar{b}-2\bar{c})$ and $(\bar{c}-2\bar{a})$ is 14 (D) $3(\bar{a}-\bar{b})$, $(3\bar{b}-\bar{c})$ and $(3\bar{c}-\bar{a})$ is 52

43. If a point P is taken on xy = 2 and then a normal is drawn from P on the ellipse $\frac{x^2}{6}+\frac{y^2}{3}=1$ which is perpendicular to $x+y=8$, then P is

(A) (1, 2) (B) (–1, –2)

(C) (2, 1) (D) (–2, –1)

44. A circle S having centre (α, β) intersect at three points A, B and C such that normals at A, B and C are concurrent at (9, 6) for parabola $y^2 = 4x$ and O is origin. Then,

(A) Sum of modulus of slopes of normals at points A, B and C is 6

(B) $\alpha+\beta=4$

(C) Magnitude of normal having negative slope is 2

(D) Circle S also passes through O

Space for Rough work

45. If $f(x) = \|x-1|+|x-3|-|2x-1\| + \|x-1|+|x-3|+|2x-1\|$, then

(A) $f(x)$ is non-differentiable at x = 1 (B) $f(x)$ is non-differentiable at $x = \frac{3}{2}$

(C) $f(x)$ has minimum value is 2 (D) x = 1 is point of local minima

46. If $\int \cos 2018x \sin^{2016} x \, dx = f(x) + c$, where $f(0) = 0$, then

(A) $\int_0^{\pi} f(x)dx = f(\pi)$ (B) $f\left(\frac{\pi}{4}\right) = \frac{1}{2^{1009}} \cdot \frac{1}{2017}$

(C) $f\left(\frac{\pi}{4}\right) = \frac{1}{(2018)2^{1009}}$ (D) $f(x)$ is odd function

47. In a ΔABC, if $a^2 - b^2 - c^2 + (x-2)bc = 0$, then x can be equal to

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$

(C) 2 (D) 4

48. If $f(x) = \{b^2 + (a-1)b + 2\}x - \int (\sin^2 x + \cos^4 x)dx$ be an increasing function of $x \in R$ and $b \in R$, then a can take value(s)

(A) 0 (B) 1

(C) 2 (D) 4

49. Let $f(x)$ be a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t)dt = x\sqrt{f(x)}$ with $f(1) = \frac{1}{2}$. Then

(A) $f(x)$ is increasing function in domain (B) $f(x)$ is decreasing function in domain

(C) $f\left(\sqrt{2}+1\right) = \frac{1}{2\sqrt{2}}$ (D) $f\left(\sqrt{2}+1\right) = \frac{1}{4}$

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If $x+y=3$ is tangent at (2, 1) on hyperbola, intersects asymptotes at A and B such that $AB=8\sqrt{2}$. If centre of hyperbola is (–1, –1), then least possible value of sum of semi-transverse axis and semi-conjugate axis is $p\sqrt{q}$, where HCF (p, q) = 1 and $\sqrt{q}$ is irrational number. Then, $(p+q)$ is equal to

51. If $x^2+y^2=16$ is director circle of locus of centre of a moving ellipse touches both the axes in first quadrant, then if major axis and minor axis are not fixed, then maximum value of product of semi-major and semi-minor axis is

52. If θ is acute angle of intersect between two curves $x^2+5y^2=6$ and $y^2=x$, then $\cot\theta$ is equal to

53. If 8 harmonic means are inserted between two numbers a and b (a < b) such that arithmetic mean of a and b is $\frac{5}{4}$ times equal to geometric mean of a and b, then $\left(\frac{H_8-a}{b-H_8}\right)$ is equal to

54. If $f(x)=\sum_{i=1}^{\infty}\frac{x}{\{(i-1)x+1\}(ix+1)}$, then $f(2017)+f\left(\frac{1}{2017}\right)$ is equal to

Space for Rough work